

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Systems for Data Science, Performance Engineering, Optimization in CS

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Go, TypeScript, JavaScript, C++, Java, SQL, Bash

LLM & Agentic Systems

LLM agents, tool use, function calling, MCP, agent orchestration, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama

Retrieval / Search / RAG

RAG, RRF, cross-encoder reranking, MMR, knapsack token budgeting, Brave Search, web scraping, sentence-transformers

Backend & Databases

FastAPI, Node.js, Express, REST APIs, async services, WebSockets, durable workflows, PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps

AWS, GCP, Azure, Docker, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production NLP/clinical models, cutting manual annotation work by **100+ hours/month** and improving labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and security across releases, hardened authentication and rate limiting, and remediated **20+ VAPT findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input, iterating on prompt design and response-quality heuristics for a real customer-facing experience while building **20+ Node.js + Fastify REST APIs** over **PostgreSQL**.

Projects

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **AI agent** that orchestrates typed retrieval planning, page rerank, deterministic tool-use extraction with **LLM fallback**, output verification, and **cell-level provenance** to convert open-ended topic queries into structured entity tables.
- Hardened the service with **150+ automated tests**, **provider routing and fallback** across OpenAI/Groq backends, pipeline instrumentation, and a reviewer-facing trust UI for validation and debuggability across LLM providers.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an AI research agent that fetches web sources, generates multi-step research digests, and **streams real-time responses to the client over WebSocket**, prototyped end-to-end on Cloudflare Workers with embedded SQL storage.
- Designed durable **workflow orchestration** via WorkflowEntrypoint and Durable Objects so multi-step agent tasks survive process restarts and tool failures, with conversation state and agent memory persisted server-side in D1 without an external database.

Spark ETL Performance Optimization - NYC TLC Taxi Dataset

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed **large-scale Spark ETL pipeline** on the NYC TLC taxi dataset by tuning broadcast joins, caching, partitioning, adaptive Spark settings, and column pruning to reduce shuffle volume and wall-clock runtime.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks and documented a repeatable optimization workflow comparing naive vs. tuned runs as a reusable reference for downstream analytics jobs.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.